

EDSGN/CSE 561 Practicum 1 – Fall '25

Product improvement for **Beats Solo 3 Wireless Headphones** based on user reviews scraped from Google reviews [1]

Team –

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Introduction

Our objective was to generate suggestions for improvement for a popular product that has publicly available reviews. We chose Solo 3 Wireless On-ear headphone from Beats for this analysis. Our solution involves identifying the aspects that led to negative, or low-star, reviews and generate recommendations based on them.

The report is structured as follows, in this section we introduce the problem, in the next we describe our overall pipeline, in the sections that follow we describe the tools we used and our design decisions. Finally, we close with the recommendations generated by our analysis.

Our solution

The following set of high-level steps are used to generate recommendations for product improvement.

1. Obtaining reviews
2. Identifying the sources of user frustration
3. Generating recommendations to eliminate them

We use a variety of techniques to identify the themes common in negative/low-star reviews that frustrate the users. In the next sections we will present the techniques used in details and describe our design choices.

Obtaining reviews

Web Scraping

Google reviews is a product of reviews collected from various sources, thus we decided to scrape it to collect a diverse set of reviews from across the internet. We used Selenium Web Driver to automate a Chrome window to search for the product review in Google search engine and then collect reviews from there. The automated window was able to simulate use actions such as clicks and scrolls. To prevent deletion of the automated nature of the window we relied on waiting for random number of seconds from 1 to 5 before each action. After automatically clicking on the “more reviews” button repeatedly, for a large number of times, we scrape the HTML of the webpage. The HTML is a tree-like structured document which can be navigated using “XPaths” which are a sequence of nodes that define the path from the root to the element of interest. The presence of ARIA attributes, that make the webpage accessible for assistive technologies such as screen readers, we were able to identify

- a) The username of the reviewer
- b) The text of the user’s review
- c) The ratings given by the user
- d) The time when the review was added

This data was picked from the HTML and dumped into a CSV file, which became the raw input for our pipeline.

Cleaning reviews

The data obtained from scraping the Google reviews site contained the text of the review, the star ratings, and when the reviews were posted. However, the ratings were of the form “Rated 3.0 out of 5.0 stars,”. We used regular expression parsing to obtain the number of stars from the ratings columns. We did not lose any rows due to parsing failures. The date of review was present in the format “posted a week ago”, or “posted 2 years ago” and so on. These were again converted to either the beginning of the corresponding year, month, week, or day using a sequence of regular expression patterns. The name of the reviewer was dropped.

Sanity checks

1. We checked that all number of stars fell between 1 and 5.
2. We checked the distribution of review text length to identify the distribution of very short reviews

```
2025-10-17 22:40:43,807 - s1_ingest - INFO - Loading raw data from data/raw/reviews.csv...
2025-10-17 22:40:43,819 - s1_ingest - INFO - Loaded 5930 rows with columns: ['review_text', 'stars', 'date']
2025-10-17 22:40:43,819 - s1_ingest - INFO - Dropped 'date' column as per spec
2025-10-17 22:40:43,819 - s1_ingest - INFO - Parsing star ratings...
2025-10-17 22:40:43,836 - s1_ingest - INFO - Valid star ratings: 100.00%
```

Figure 1: Output from the ingestion phase that parses the csv obtained after web scraping

Identifying the sources of user frustration

The pain points that users feel can be inferred from either the star ratings they have given or the sentiment of their reviews. We chose to explore both the aspects. While number of stars given by the user was available we used pre-built model to infer the sentiment of the review.

Sentiment analysis was done using the following models

1. VADER[2] – a rule-based model for sentiment analysis that scores the text for positive, negative, or neutral sentiments, as well as returns a composite score
2. SiEBERT[3] – a 400 million parameters deep-learning model for sentiment analysis that classifies text as positive, negative, or neutral

We used the composite score from Vader to classify sentiment of a review as negative if it were less than -0.05, neutral if it is between -0.05 and 0.05, and as positive if it is higher than 0.05. We chose VADER in favour of SiEBERT because the accuracy gains from using SiEBERT were approximately 5% while the resources required to run SiBERT were considerably higher.

What the user is talking about in a review can be identified either using n-grams, or a rule-based aspect identification.

1. n-gram based approach – bi-grams are set of two consecutive words that appear in a set of documents. A document can just be one review as well. By analysing the frequency of bi-grams we can find out which pair of words appear most frequently together. Most of the user complaints can be identified using two words.
2. aspect based approach – each product has a set of well-known aspects such as for a headphone people care about the battery life, and the comfort. A user can talk about an aspect using any of the well-known keywords, such as battery life can be talked of using the word “charging”, or “usage”. Therefore, we define a set of aspects for the headphones, and for each aspect we define a list of keywords that may talk about it.

What the user is talking about, generated by these approaches, was then corroborated by running a k-means cluster analysis to verify that the information we have coded matches the information mined from the review text.

Finally, to identify what the user is talking about, that is causing them pain we examine the following four pairs

1. bi-grams in low rating reviews
2. bi-grams in negative sentiment reviews
3. rule-based aspects in low rating reviews
4. rule based aspects in negative sentiment reviews

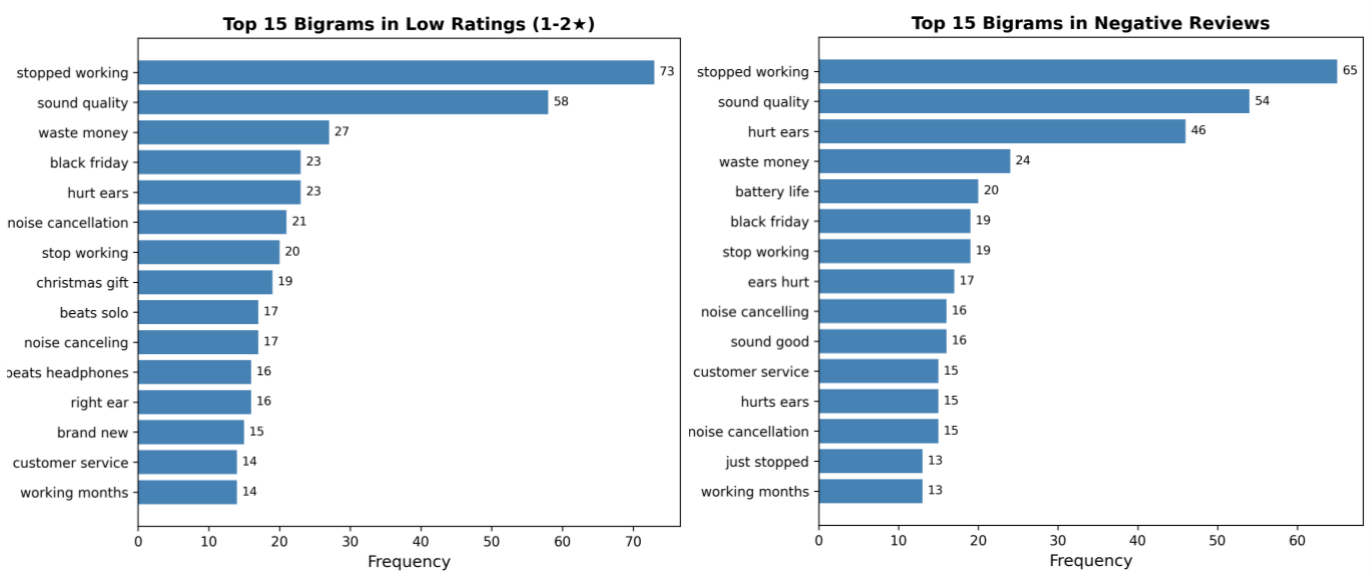


Figure 2: Bi-grams (2-word) frequency for reviews with low ratings and reviews with negative sentiment

	aspect	num_mentions	pct_total	avg_stars	low_star_share	neg_sent_share
5	Controls/Features	355	5.994596	3.946479	27.887324	15.492958
6	Price/Value	1208	20.398514	4.117550	23.509934	13.245033
3	Connectivity/Bluetooth	1057	17.848700	4.105960	22.800378	12.866604
8	App/Software	40	0.675447	4.000000	27.500000	12.500000
1	Comfort/Fit	2218	37.453563	4.287196	18.349865	10.369702
0	Sound/Audio Quality	3756	63.424519	4.427050	14.563365	7.720980
2	Battery/Life	1606	27.119216	4.390411	15.193026	7.471980
4	Build/Design	1172	19.790611	4.482082	12.883959	6.143345
7	Case/Portability	388	6.551841	4.577320	8.762887	3.865979

Figure 3: Percentage of low-rating reviews and reviews with negative sentiment that mention an aspect

From figures 2 and 3, we see that low ratings, and negative sentiment reviews go hand in hand. But, rule-based aspects seem to address topics such as Controls/Features, that may not be captured by Bi-grams. Therefore, we chose to pick aspects with largest share of negative review sentiments for generating recommendations.

From figures 2, and 3, we pick the following pain points to address –

1. Controls/features (using aspects) – 355 reviews mentioned this aspect
2. Hurts ears (using bi-grams) – 64 reviews had related bigrams and had negative sentiment

Generating recommendations

Once we have identified what aspect of the product must be addressed, we take all the reviews that are negative and talk about that aspect, and feed them into ChatGPT after embedding them in the following prompt

```
“
As a product design expert, with a consumer first attitude,
1. look at the following reviews of Beats Solo 3 Wireless Headphones,
2. perform a frequency analysis of complaints regarding {{aspect}},
3. examine the complains based on pareto principles - 20% improvements giving 80% gains, and finally
4. suggest 2 improvements based on the reviews below, enclosed within triple back ticks

...
{{selected reviews}}
...
”
```

Final recommendations

Based on the pipeline described above we make the following recommendations to improve Beats Solo 3 Wireless On-ear headphones

1. ‘Add a "Loudness / Outdoor mode" toggle in firmware.’ which will directly enhances perceived value and competitiveness vs. Bose/Sony. [4]
2. ‘Introduce multi-device Bluetooth switching (e.g., between laptop and phone).’ which will bring Solo 3 in line with user expectations in 2025, resolving most “outdated control” complaints. [4]
3. ‘Replace static metal hinge with *flex-adjust* polymer/dual-spring arms that maintain consistent acoustic seal at ≤ 2 N side pressure.’ which improves comfort for varied head sizes [5]

[1] Google Inc., (2025) Google search result for “Beats solo 3 wireless on ear headphones”

<https://www.google.com/search?q=Beats+solo+3+wireless+on+ear+headphones>

[2] Hutto, C.J. & Gilbert, E.E. (2014). VADER: A Parsimonious Rule-based Model for Sentiment Analysis of Social Media Text. Eighth International Conference on Weblogs and Social Media (ICWSM-14). Ann Arbor, MI, June 2014.

[3] Hartmann, J., Heitmann, M., Siebert, C., & Schamp, C. (2023). More than a Feeling: Accuracy and Application of Sentiment Analysis. International Journal of Research in Marketing, 40(1), 75–87.

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<https://www.sciencedirect.com/science/article/pii/S0167811622000477>

[4] Kumar, R., Wagle, A., Misra V. (2025). ChatGPT conversation on multilingual sentiment analysis pipeline [Online discussion]. OpenAI ChatGPT. Retrieved from <https://chatgpt.com/share/68f3048d-7dcc-800e-944c-d219ccd30705>

[5] Kumar, R., Wagle, A., Misra V. (2025). ChatGPT conversation on multilingual sentiment analysis pipeline [Online discussion]. OpenAI ChatGPT. Retrieved from <https://chatgpt.com/share/68f304d0-f5a4-800e-871e-dd9bb39f5f4e>